Topic 3

Production, uses and risks of ionising radiation from radioactive sources

3.1 Evaluate the social and ethical issues relating to the use of radioactive techniques in medical physics

3.2 Describe the properties of alpha, beta, gamma, positron and neutron radiation

3.3 Recall the relative masses and relative electric charges of protons, neutrons, electrons and positrons

3.4 Recall that in an atom the number of protons equals the number of electrons

3.5 Describe the process of $\beta-$ decay (a neutron becomes a proton plus an electron)

3.6 **Describe the process of $\beta+$ decay (a proton becomes a neutron plus a positron)**

3.7 Explain the effects on the atomic (proton) number and mass (nucleon) number of radioactive decays (α, β and γ decay)

3.8 **Use given data to balance nuclear equations**

3.9 **Describe the features of the *N-Z* curve for stable isotopes**

3.10 **Identify isotopes as radioactive from their position relative to the stability curve**